📘 Assignment 4 – EV Charging Station Scheduling

🔷 Problem Statement

A charging station has K ports with different speeds and costs. Each vehicle i is defined by:

- Arrival time: a_i
- Departure time: d_i
- Required charge time (slowest port): c_i

If assigned to port k, charging time becomes:

$$t_{ik} = \left\lceil \frac{c_i}{k} \right\rceil$$

Each vehicle must:

- Be assigned to **exactly one port**
- Charge **continuously (non-preemptive)**
- Finish before departure

👉 Objective: Find a **minimum-cost feasible assignment**

🔷 Goal

Minimize total charging cost:

$$\min Z = \sum_{i=1}^{N} \sum_{k=1}^{K} X_{i,k} \cdot (t_{ik} \cdot p_k)$$

Where:

- $X_{i,k} = 1$ if vehicle i assigned to port k, else 0 ↓

🔷 State Space

All possible assignments of vehicles to ports:

$$\mathcal{S} = \{(x_1, x_2, \ldots, x_N) \mid x_i \in \{1, 2, \ldots, K\}\}$$

In your code (important restriction):

$$x_i \neq x_j \quad \forall i \neq j$$

👉 So actual state space:

$$\text{All permutations of assigning } N \text{ vehicles to } K \text{ distinct ports}$$

🔷 Decision Variables

✔ **In mathematical form:**

$$X_{i,k} = \begin{cases} 1 & \text{if vehicle } i \text{ assigned to port } k \\ 0 & \text{otherwise} \end{cases}$$

OR (your code representation):

$$x_i \in \{1, 2, \ldots, K\}$$

Where:

- x_i = port assigned to vehicle i

🔷 Constraints

1️⃣ Each vehicle assigned exactly one port

$$\sum_{k=1}^{K} X_{i,k} = 1 \quad \forall i$$

(Code equivalent: one `port[i]` per vehicle)

2️⃣ Port uniqueness (your code assumption)

$$x_i \neq x_j \quad \forall i \neq j$$

👉 Means:

- One vehicle per port (no sharing)

3️⃣ Arrival constraint

$$s_i \geq a_i$$

(Implicit in your code: charging starts at arrival)

4️⃣ Completion before departure

$$a_i + t_{ik} \leq d_i \quad \text{if } x_i = k$$

Where:

$$t_{ik} = \left\lceil \frac{c_i}{k} \right\rceil$$

(Code:)

```cpp
if (arrival + time_needed <= departure)
```

5️⃣ Charging time definition

$$t_{ik} = \left\lceil \frac{c_i}{k} \right\rceil$$

(Code:)

```cpp
time_needed = (charge + k - 1) / k;
```

6️⃣ Domain constraint

$$1 \leq x_i \leq K$$

⚠️ Important Modeling Assumption (from your code)

$$N \leq K$$

Otherwise:

$$\text{No feasible solution}$$

🔷 Objective Function (Expanded Form)

$$Z = \sum_{i=1}^{N} \sum_{k=1}^{K} X_{i,k} \cdot \left(\left\lceil \frac{c_i}{k} \right\rceil \cdot p_k \right)$$

🔷 Cost Computation

For each vehicle-port pair:

$$\text{Cost}_{ik} = \left\lceil \frac{c_i}{k} \right\rceil \cdot p_k$$

🔷 Heuristics (Conceptual + Practical)

Even though your Z3 model is exact, these heuristics explain intuition:

🔹 1. Earliest Deadline First (EDF)

$$\text{Sort vehicles by } d_i$$

👉 Vehicles with tight deadlines get priority

🔹 2. Cheapest Feasible Port First

$$\min_k \{p_k \mid a_i + t_{ik} \leq d_i\}$$

👉 Choose lowest-cost valid port

🔹 3. Minimum Slack Time

$$\text{Slack}_i = d_i - a_i - t_{ik}$$

👉 Smaller slack → higher urgency

🔹 4. Lower Bound Estimation

$$Z \geq \sum_{i=1}^{N} \min_k (t_{ik} \cdot p_k)$$

👉 Helps pruning (used implicitly by Z3)

📘 Assignment 5 – Short-Term Course Scheduling (SAT-Based)

🔷 Component: Problem Statement

There are M rooms and N short-term courses. Each course i has:

- Preferred start day: s_i
- Deadline: d_i
- Duration: t_i

Each course must:

- Run on consecutive days
- Be assigned to exactly one room
- Finish within its deadline

If a course starts at day t, it occupies:

$$[t,\ t + t_i - 1]$$

🔷 Component: Goal

Determine whether a feasible schedule exists such that:

$$\text{All courses are scheduled without conflict and within deadlines}$$

👉 This is a SAT (satisfiability) problem, not optimization.

🔷 Component: State Space

All possible assignments of:

- Course → Room
- Course → Start Time

$$\mathcal{S} = \{(r_i, s_i') \mid r_i \in \{1, ..., M\},\ s_i' \in [s_i, d_i - t_i + 1]\}$$

Also includes:

- Time intervals occupied in each room

🔷 Component: Decision Variables

🔹 Option 1: $z_{i,j,t}$

$$z_{i,j,t} = \begin{cases} 1 & \text{if course } i \text{ starts at day } t \text{ in room } j \\ 0 & \text{otherwise} \end{cases}$$

🔹 Option 2: $x_{i,j},\ y_{i,t}$

$$x_{i,j} = \begin{cases} 1 & \text{if course } i \text{ assigned to room } j \\ 0 & \text{otherwise} \end{cases}$$

$$y_{i,t} = \begin{cases} 1 & \text{if course } i \text{ starts at day } t \\ 0 & \text{otherwise} \end{cases}$$

🔷 Component: Constraints

1️⃣ Each course assigned exactly one room

$$\sum_{j=1}^{M} x_{i,j} = 1 \quad \forall i$$

(Option-1 equivalent:)

$$\sum_{j=1}^{M} \sum_{t} z_{i,j,t} = 1$$

2️⃣ Each course starts exactly once

$$\sum_{t=s_i}^{d_i - t_i + 1} y_{i,t} = 1 \quad \forall i$$

3️⃣ Valid start window

$$s_i \leq t \leq d_i - t_i + 1$$

Component: Heuristics

Earliest Deadline First (EDF)

$$\text{Sort courses by } d_i$$

Minimum Slack Time

$$\text{Slack}_i = d_i - s_i - t_i$$

Earliest Feasible Start

$$t = \min\{t \mid s_i \leq t \leq d_i - t_i + 1\}$$

Conflict Minimization

Assign rooms to reduce overlapping intervals